Interpreting Inequalities

Exam code: 0607

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Interpreting inequalities

What is an inequality?

- An **inequality** tells you that something is **greater than** (>) or **less than** (<) something else
  - $x > 5$ means x is greater than 5
    - x could be 6, 7, 8, 9, ...
- Inequalities may also include being **equal** (=)
  - ⩾ means **greater than or equal to**
  - ⩽ means **less than or equal to**
    - $x \leqslant 10$ means x is less than or equal to 10
      - x could be 10, 9, 8, 7, 6,....
- When they **cannot** be **equal**, they are called **strict** inequalities
  - > and < are **strict** inequalities
    - $x > 5$ **does not** include 5 (strict)
    - $x \geqslant 5$ **does** include 5 (not strict)

How do I find integers that satisfy inequalities?

- You may be given **two end points** and have to list the **integer** (whole number) values of x that **satisfy** the inequality
  - Look at whether each end point is **included** or not
    - $3 \leqslant x \leqslant 6$
      - $x = 3, 4, 5, 6$
    - $3 \leqslant x < 6$
      - $x = 3, 4, 5$
    - $3 < x \leqslant 6$
      - $x = 4, 5, 6$
    - $3 < x < 6$
      - $x = 4, 5$
- If only **one end point** is given, there are an infinite number of integers
  - $x > 2$
    - $x = 3, 4, 5, 6, ...$
  - $x \leqslant 2$
    - $x = 2, 1, 0, -1, -2, ...$

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

then just list $x = 2, 1$

- You may be asked to find integers that satisfy **two** inequalities
  - $0 < x < 5$ and $x \geqslant 3$
    - List **separately**: $x = 1, 2, 3, 4$ and $x = 3, 4, 5, 6, \ldots$
    - Find the values that appear in **both** lists: $x = 3, 4$
- If the question does not say x is an integer, do **not assume** x is an integer!
  - $x > 3$ actually means **any value** greater than 3
    - 3.1 is possible
    - $\pi = 3.14159\ldots$ is possible
- You may be asked to find the **smallest** or **largest** integer
  - The smallest integer that satisfies $x > 6.5$ is 7

Worked Example

List all the integer values of x that satisfy

$$-4 \leq x < 2$$

Answer:

Integer values are whole numbers
$-4 \leq x$ shows that x includes -4, so this is the first integer

$$x = -4$$

$x < 2$ shows that x does not include 2
Therefore the last integer is $x = 1$

$$x = 1$$

For the answer, list all the integers from -4 to 1
Remember integers can be zero and negative

$$x = -4, -3, -2, -1, 0, 1$$

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